Schema2Bench: Evidence-Grounded Cross-Domain Text-to-SQL Benchmark Migration

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***Abstract*—Text-to-SQL benchmarks compare models that answer natural-language questions over relational databases. Reusing their query structures can extend evaluation to new databases without constructing every task from scratch. However, executable SQL can misrepresent the intended task, and changes in workload composition can alter model comparisons. Structural coverage alone does not establish whether a migrated benchmark preserves source behavior or reflects model differences on the target database. In this paper, we propose Schema2Bench (S2B), a framework for cross-domain benchmark migration. Benchmark Construction grounds source query patterns in target schemas and sampled values, then generates questions from SQL. Workload Selection adjusts composition under fixed quotas, bounds fidelity changes and identifies structural deficits in the candidate pool. Benchmark Evaluation separately measures source fidelity and agreement with model accuracy gaps on target references. We compare five construction methods on eight BIRD databases with ten models, 2,000 tasks per method and 906 target references. In the main comparison under Verified execution scoring and declared weights, S2B reduces the template baseline's target gap from 0.225 to 0.169 and improves source fidelity over all four comparators. Fixed-SQL question rewriting lowers target gap but worsens source fidelity, showing why benchmark continuity and target model selection require separate assessment.**

I. Introduction

Text-to-SQL systems let users query relational databases through natural language. Recent methods improve SQL generation through prompting, specialized models and feedback [1–3]. Their evaluation relies on workloads such as Spider and BIRD [4, 5], whose questions refer to particular tables, relationships and values. Choosing a model for another database requires tasks grounded in that database. Reusing source query structures can reduce construction effort, but the resulting workload must also be assessed as a measurement instrument.

Existing approaches address different parts of this problem. RingSQL uses schema-independent templates to produce training pairs, while OmniSQL synthesizes diverse databases, queries and questions for training [6, 7]. SQLMorph expands joins and perturbs questions to diagnose model weaknesses [8]. These methods establish useful construction and diagnostic techniques. Their generation quality, structural coverage or downstream training gains do not, however, determine whether a migrated workload preserves source model comparisons or supports model selection on the target database. We study this distinction in *cross-domain benchmark migration*.

Challenges. *First, executable SQL does not establish task meaning.* Execution-based evaluation can miss semantic differences [9], and data-grounded explanations show why database contents matter when checking SQL [3]. When transferring a query pattern, matching column types alone can change the entities, relationships or measures being queried. Value-linking studies further show that syntactically valid queries can use the wrong database values [10]. Benchmark construction therefore needs to ground both SQL and its question in the target schema and data.

An illustrative example. Figure 1 asks which customers placed at least two orders. On source order rows, `COUNT(*) >= 2` works. On target order lines, C1 has two rows for one order; C2 has two orders. The transferred pattern returns both, whereas `COUNT(DISTINCT oid)` correctly retains C2. Executability does not distinguish counting units.

Second, a valid workload can change model comparisons. Diagnostic perturbations expose sensitivity to query structure and wording [8, 11]; benchmark-design and concurrence studies also distinguish evaluation claims from the evidence needed to support them [12, 13]. Matching structural frequencies alone need not preserve model behavior. Selection can adjust frequencies only within the available candidate pool, and source fidelity need not imply agreement with target references.

Suppose models A/B score 75%/50% on the source but 50%/75% on target references. Reproducing source rates preserves A's advantage; reproducing target rates favors B. These hypothetical scores separate benchmark continuity from target model choice. Compact evaluation methods estimate performance on an existing benchmark [14]; migration additionally changes the database and task population.

Our methodology. Schema2Bench (S2B) contains three modules (Fig. 2). *Benchmark Construction* instantiates source query patterns using target schemas and sampled values, then generates questions from SQL. *Workload Selection* adjusts structural composition under fixed quotas, with a tight bound on source-fidelity change and a coverage bound for deficits that selection cannot remove. *Benchmark Evaluation* measures source fidelity and target alignment separately, retaining unknown scores in direct method comparisons. Candidate screening uses earlier outputs; selection itself reads no evaluator responses. Execution and consistency checks screen operational failures without certifying task semantics.

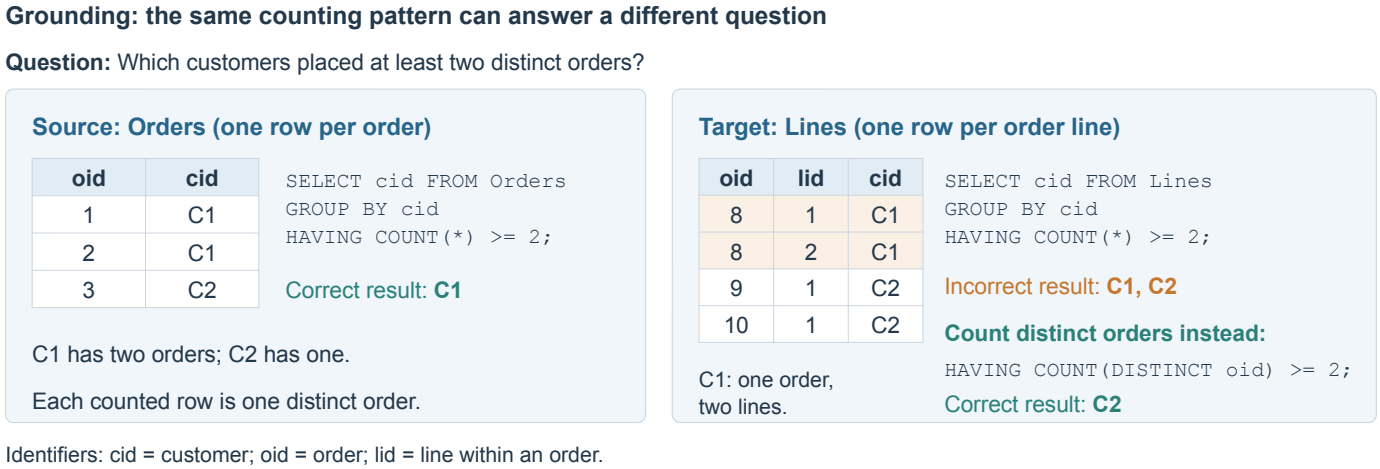


Fig. 1. An illustrative grounding error: transferring `COUNT(*)` from orders to order lines changes what is counted. Both target queries execute, but only distinct-order counting answers the question. Data are constructed for explanation.

Experimental evaluation. We compare five methods on eight BIRD databases with ten models. S2B improves source fidelity under Verified scoring, while external adaptations have lower target gap. Construction accounts for most improvement over Template; changing question wording improves target gap but worsens source fidelity. These results show why generation quality and target model selection require separate evaluation.

Contributions. We make three contributions:

1) **Benchmark Construction.** We construct target tasks from source SQL patterns using schema and value evidence, with explicit question–SQL alignment and failure handling (Section IV-A).
2) **Workload Selection.** We bound the fidelity change from quota-constrained selection and characterize structural deficits in the candidate pool (Section IV-B).
3) **Benchmark Evaluation.** We distinguish source fidelity from target model choice and evaluate construction, selection and wording effects under execution and semantic uncertainty (Sections IV-C and V).

II. Preliminaries

A. Text-to-SQL Tasks

A domain is a database with a fixed schema and data snapshot. Task (d_i, x_i, s_i) pairs question x_i and reference SQL s_i on domain d_i. Migration (Fig. 2) constructs a release of N tasks across D domains, using source SQL queries $\mathcal{S}$ for patterns and statistics. Separate workload $\mathcal{W}_s$ supplies the behavioral reference for $M \geq 2$ models.

B. Evaluation Objectives

Source fidelity measures continuity with source model behavior; target alignment measures agreement with target-reference model differences. Rankings and regret assess model selection (Section IV-C1).

III. Schema2Bench Overview

In this section, we first formulate benchmark migration in Section III-A, then outline Schema2Bench in Section III-B. Its three modules are *Benchmark Construction*, *Workload Selection* and *Benchmark Evaluation*. Sections IV-A, IV-B and IV-C describe their implementations, respectively.

A. Problem Formulation

Candidate pool $\mathcal{C}$ retains IDs and evidence. Membership $z \in \{0,1\}^{|\mathcal{C}|}$ specifies release $\mathcal{W}_z$, selecting item i when $z_i = 1$. Strata partition candidates within domains; $A_{g,i}$ indicates membership in stratum g, with quota q_g. Feasible memberships $\mathcal{F}$ satisfy $Az = q$ and $\mathbf{1}^\top z = N$, where $\mathbf{1}$ is the all-ones vector. Quotas fix domain sizes $N_d > 0$ and require sufficient candidates per stratum.

For a sampled initial release $z^0 \in \mathcal{F}$, selection seeks

$$\min_{z \in \mathcal{F}} L_{\text{struct}}(z) \quad \text{subject to } H(z, z^0) \leq \tau. \tag{1}$$

L_{struct} measures source-structure mismatch. H bounds changes in source fidelity error (TEF), with tolerance τ (Section IV-B), not absolute loss.

B. Framework Overview

Figure 2 connects the three modules: construction produces candidate pairs, selection controls workload composition, and evaluation compares models on source, selected and reference tasks.

IV. Methodologies

We detail the three modules below.

A. Benchmark Construction

Schema2Bench instantiates source operator patterns with target tables, types, keys and sampled values, then derives questions from SQL. The evaluated implementation is programmatic; Section V-C3 tests LLM question generation.

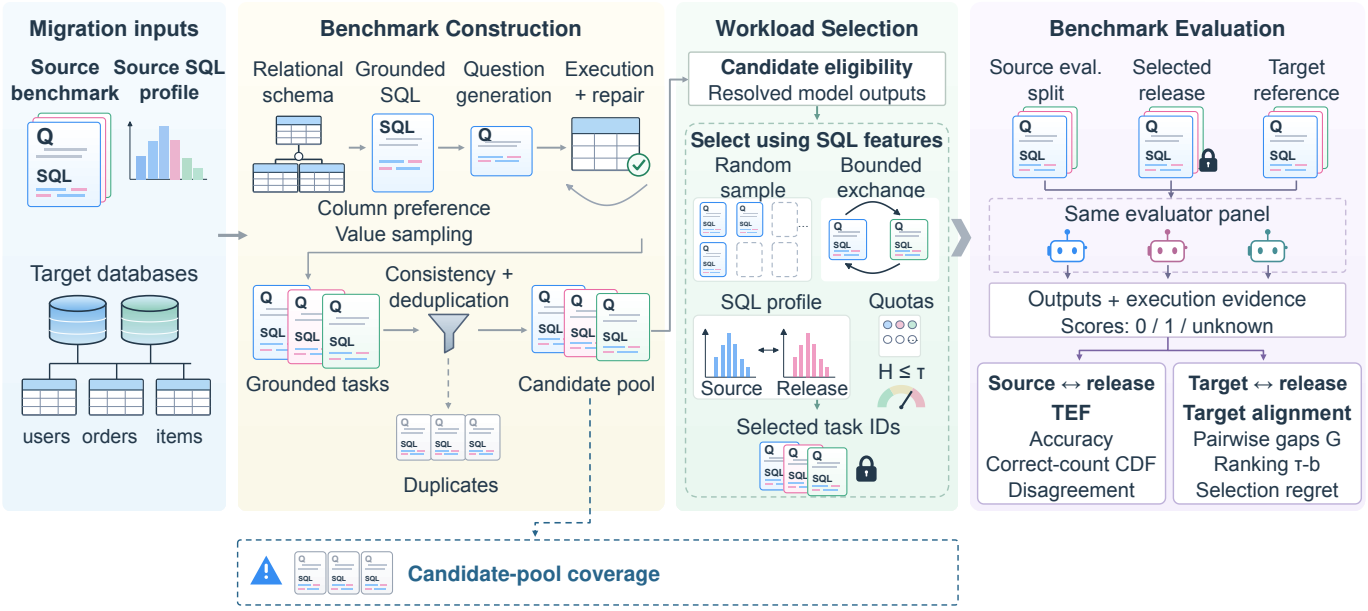


Fig. 2. The three Schema2Bench modules: Benchmark Construction, Workload Selection and Benchmark Evaluation. Q: question; CDF: cumulative distribution function; H: fidelity-change bound; τ: tolerance; G: target gap; τ-b: Kendall rank correlation.

Binding operators to the database. Columns whose names suggest identifiers are excluded from measures and descriptive projections. Text and name/title/description cues guide projected columns; categorical cues guide grouping. SUM requires cues that the column represents an additive measure. Join candidates follow declared keys or entity-key heuristics. Equality predicates use observed values, with up to fifty samples per column. If a join cannot be built, the generator falls back to a grounded filter or scalar aggregate. Structural features are recomputed from the final SQL; the recorded construction stratum is retained separately.

Keeping the question aligned with SQL. Questions name columns, filters, join relationships and aggregation roles. For recognized joins, `COUNT(*)` describes matched rows: order-line rows in Fig. 1. S2B does not infer a general entity-preserving `COUNT(DISTINCT)` repair; distinct-order counting requires different SQL. Relationship clarification also applies to Template.

Admission and failure handling. Execution checks and bounded repair handle operational failures. A final LIMIT clause requires recoverable ordering over projected columns or proof of a unique cutoff; the question must state the ordering and limit. Consistency checks screen literal, entity and ordering mismatches; deduplication removes repeated tasks. These checks reduce identifiable failure modes without certifying question–SQL meaning. Canonical IDs pair Template/S2B alternatives without assuming identical semantics (Section V-D2).

B. Workload Selection

Candidate eligibility. Definite execution scores and normal termination determine $\mathcal{C}$ (Section V-A). This screen depends on outputs; selection reads only SQL features, schema evidence, metadata and a seed. Main-study strata combine domain and recorded construction group.

Selection using SQL features. Sampling q_g candidates per stratum gives initial release z^0. Exchanges improve structural matching while limiting their possible effect on the measured response profile. They target Eq. (1) while preserving quotas. Structural loss averages operator-event and SQL-only difficulty total variation (TV), six shape deviations, relative mean join-count and SQL-length errors:

$$L_{\text{struct}}(z) = \frac{1}{5}\left(\frac{\|\widehat{P}_o - P_o^*\|_1}{2} + \frac{\|\widehat{P}_d - P_d^*\|_1}{2} + \frac{\|\widehat{P}_h - P_h^*\|_1}{6} + \frac{|\widehat{j} - j^*|}{j^*} + \frac{|\widehat{\ell} - \ell^*|}{\ell^*}\right). \tag{2}$$

P_o, P_d are operator-event and difficulty distributions; P_h contains filtering/join/grouping/aggregation/ordering/LIMIT rates. j, ℓ are mean joins and SQL length in whitespace-separated tokens. Hats/stars denote release/source statistics; $\|\cdot\|_1$ sums absolute coordinates. Difficulty and shapes count tasks. Source means are positive; relative errors can exceed one.

SQL-only difficulty sums joins, twice the additional SELECT and set-operator counts (UNION/INTERSECT/EXCEPT), and indicators for grouping, HAVING, aggregation, ordering, CASE, DISTINCT and at least two AND/OR tokens. Scores ≤ 1, 2–3 and ≥ 4 define easy, medium and hard, differing from Spider labels and behavioral/semantic difficulty.

Let r_d count tasks removed from initial release z^0 in domain d. The budget function $H = H(z, z^0)$ is

$$H = \sum_d (w_d^A + w_d^R)\frac{r_d}{N_d} = \frac{r}{N} + \frac{1}{D}\sum_d \frac{r_d}{N_d}, \qquad r = \sum_d r_d, \tag{3}$$

where $w_d^A = 1/D$ weights accuracy and $w_d^R = N_d/N$ weights the correct-count cumulative distribution function (CDF) and disagreement across domains (Section IV-C1). Each exchange e replaces an initial item with an unused candidate outside the initial release in the same stratum. For resulting membership z', gain is $\Delta_e = L_{\text{struct}}(z) - L_{\text{struct}}(z')$ and cost is $c_d = 1/N + 1/(DN_d)$. Choose the largest positive Δ_e/c_d fitting the budget, breaking ties by resulting loss and IDs. Update z, H; stop at no admissible improvement or the step cap. Feasibility and decreasing structural loss hold, without global optimality.

For fixed source moments, joint response laws and domain counts, $|L_{\text{TEF}}(z) - L_{\text{TEF}}(z^0)| \leq H$. Response vectors at Hamming distance h_H change accuracy by at most h_H/M, and CDF plus half-weighted disagreement by at most $\frac{h_H}{M} + \frac{h_H(M-h_H)}{M(M-1)} \leq 1$. Coupling removed/added items gives Eq. (3), attained by all-zero source and initial responses with all-one replacements. This perturbation bound also holds in expectation under fixed joint laws, including deterministic saved responses.

1) Candidate Coverage: Over SQL types T, such as easy/medium/hard difficulty, let p and q_z denote source and release proportions, distinct from quotas q. The convex hull $\mathcal{Q} = \text{conv}\{q_z : z \in \mathcal{F}\}$ contains their feasible mixtures v. With dual type weights h, the minimum type mismatch is

$$\delta_{\text{sup}} = \min_{v \in \mathcal{Q}} \text{TV}(p, v) = \max_{h \in [0,1]^{|T|}} \left(h^\top p - \max_{z \in \mathcal{F}} h^\top q_z \right). \quad (4)$$

The coverage lower bound δ_{sup} uses a bounded-count linear program (LP) relaxation and sorted dual weights. Omitting $H \leq \tau$ preserves validity for budgeted selection. An attaining integer release certifies optimal type matching; otherwise integrality may leave a gap. With unchanged feasible quotas, adding b eligible candidates reduces TV over pooled tasks by at most b/N: replace new selections with unused old items in the same strata. This bounds structural capacity, not TEF (Section V-C4); Section V-C5 tests acquisition.

C. Benchmark Evaluation

The same M models answer source, release and reference tasks. Evaluation separates source fidelity from target alignment and retains unresolved scores.

1) Evaluation Measures: **Source fidelity (TEF).** $Y_i \in \{0,1\}^M$ contains task i's execution-match indicators, with m, n indexing models. TEF compares accuracy, correct-count CDF indicators $\mathbf{1}\{\sum_m Y_{i,m} \leq k\}$, $k = 0, \ldots, M-1$, and disagreement $|Y_{i,m} - Y_{i,n}|$. Here $\mathbf{1}\{\cdot\}$ is one when true, zero otherwise. With source/release feature means Ψ_s, Ψ_z,

$$L_{\text{TEF}}(z) = \sum_\nu w_\nu |\Psi_{z,\nu} - \Psi_{s,\nu}| = D_{\text{diff}} + D_{\text{acc}} + \tfrac{1}{2} D_{\text{dis}}. \quad (5)$$

ν indexes coordinates with weights w_ν: $1/M$ for accuracy/CDF and $1/(2\binom{M}{2})$ for disagreement. $D_{\text{diff}}, D_{\text{acc}}, D_{\text{dis}}$ are mean absolute differences in CDF, accuracy and disagreement, respectively. For selected domain items $z_d = \{i : z_i = 1, d_i = d\}$, accuracy coordinate $\Psi^A_{z,m}$ averages databases equally:

$$\Psi^A_{z,m} = \frac{1}{D} \sum_d \frac{1}{N_d} \sum_{i \in z_d} Y_{i,m}. \quad (6)$$

CDF and disagreement pool tasks; source accuracy averages 20 databases. Equal target quotas do not remove unequal source sizes. Block weights $(1, 1, 1/2)$ define the convention; Section V-D4 varies these priorities over the nonnegative weight region without changing the structural objective.

Target alignment. For construction c, let $a^r_{d,m}$ and $a^c_{d,m}$ be model m's execution-match rates on reference and constructed tasks in domain d. Target gap measures differences between their pairwise accuracy gaps:

$$G(c) = \frac{1}{D} \sum_d \frac{1}{\binom{M}{2}} \sum_{m<n} \left| (a^r_{d,m} - a^r_{d,n}) - (a^c_{d,m} - a^c_{d,n}) \right|. \quad (7)$$

Lower TEF and gap are better. For constructions a, b, $\Delta G(a,b) = G(a) - G(b)$ and ΔTEF use first-minus-second losses; negative favors the first. $I_G = G(\text{Template}) - G(\text{S2B})$ reports positive improvement. Multiplying by 100 gives gap percentage points (pp), distinct from accuracy changes. Domain-mean Kendall τ-b measures ranking with ties; regret is target accuracy lost by choosing a release winner. Common accuracy shifts preserve gaps, while near ties can reverse rankings.

2) Execution Scoring: Base scoring compares result values and column counts, including empty-result widths. Non-stop or unresolved outputs receive $[0, 1]$; parser rejection or unordered LIMIT alone does not prove error. Neither scoring layer certifies question–SQL meaning.

The Verified layer resolves unknowns using SQLite errors, row counts, unique ordering and checks of possible LIMIT results tied to original outputs (Table II).

3) Comparisons with Unknown Scores: Unknown scores induce feature intervals. Direct contrasts share source/reference uncertainty, with target outcomes separate unless prompts and outputs are identical (Section V-C2). Coordinate relaxation gives conservative outer bounds, not confidence intervals or necessarily attainable endpoints. We do not subtract independently optimized row bounds.

Bootstrap intervals condition on stated populations, weights and units; shared models/pools make pairs and releases dependent.

Lower/upper score bounds $[l_{i,m}, u_{i,m}]$ are $[0,0]$, $[1,1]$ or $[0,1]$ for mismatch, match or unknown. Weighted means bound accuracy; CDF contributions lie between $\mathbf{1}\{\sum_m u_{i,m} \leq k\}$ and $\mathbf{1}\{\sum_m l_{i,m} \leq k\}$. Disagreement is fixed when both scores are fixed, otherwise relaxed to $[0,1]$.

For method-feature intervals $A = [a_-, a_+]$, $B = [b_-, b_+]$, the minimum/maximum absolute distances are

$$d_-(A,B) = \max(a_- - b_+, b_- - a_+, 0), \quad (8)$$

$$d_+(A,B) = \max(|a_- - b_+|, |a_+ - b_-|). \quad (9)$$

For shared source-feature interval S, the discrepancy difference (first minus second) has lower/upper bounds

$$\underline{\Delta} = \min_{s \in S} \{d_-(\{s\}, A) - d_+(\{s\}, B)\}, \quad (10)$$

$$\overline{\Delta} = \max_{s \in S} \{d_+(\{s\}, A) - d_-(\{s\}, B)\}. \quad (11)$$

Both terms share source value $s \in S$. We sum weighted coordinate bounds, evaluating endpoints and piecewise-linear breakpoints. Gap contrasts similarly share signed target-reference model gaps.

V. Experimental Results

In this section, we present the experimental setup (Section V-A), main comparisons (Section V-B), and ablations of Benchmark Construction and Workload Selection (Section V-C). We then examine robustness of Benchmark Evaluation (Section V-D), generalization (Section V-E) and implications (Section V-F).

A. Experimental Setup

1) Datasets and Models: The primary comparison uses eight databases from BIRD (Table I), ten models, 2,000 tasks/method and 906 shared references. Spider supplies 7,000 training SQLs for the source profile and 1,034 development items from 20 databases for evaluation. Prompts use data definition language (DDL) schemas from retained snapshots.

The panel comprises Qwen3-4B (Instruct-2507), Granite-3.3-8B-Instruct [15], Qwen2.5-Coder-7B-Instruct [16], Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct [17], Mistral-7B-Instruct-v0.3 [18, 19], Qwen2.5-14B-Instruct, Qwen3-8B, Gemma-2-9B-it [20] and Phi-4 [21]. Qwen2.5-7B also participates in generation [22].

a) Initial qualification result: The original workload improved in 5/8 domains, failing the preregistered six-domain requirement. Later Qwen3-8B non-thinking reruns [23] and scoring changes do not revise this failure.

b) Candidate screening and selection: Both pair members require normal termination and definite ten-model scores, including compilation and row-count evidence. Incorrect responses qualify; unresolved or truncated ones do not. Exclusions chiefly concern timeouts and unprojected ordering keys (Table I); overlapping reasons exclude earlier generation failures.

The 3,778-ID pool supplies 250 pairs/domain using proportional largest-remainder quotas and within-stratum sampling. S2B features select both arms' IDs before cached scores are read; Template is not independently optimized. Equal quotas and $H \leq 0.01$ allow ten equal-cost swaps, precluding identification of normalization benefits. Ten releases overlap; scoring layers share membership. Outcome-dependent eligibility and unequal precision remain limitations.

2) Baselines: Template/S2B share the paired pool; grounding differs (Section III). RingSQL-style combines template SQL with Qwen questions. Omni reuses SQL/question-generation components with local Qwen. Synth uses the author's SQL synthesis, question generation, backtranslation and self-comparison workflow with local Qwen, removing its 25-row cap.[1] These local adaptations are not full author-system reproductions. Release sizes and evaluation populations are fixed; generation budgets differ (Table VII).

[1]SynthSQL toolkit: https://github.com/NeginKhalifat/SQL-benchmark, retained commit `dff8c75e`. It is distinct from SING-SQL and SynSQL.

TABLE I
Available → retained candidates and references after ten-model eligibility screening. Each release selects 250 pairs per domain.

Domain	Candidate pairs	Reference items
California schools	703→703	85→55
Card games	460→453	171→154
Codebase community	373→356	176→153
European football	613→601	123→91
Financial	383→382	101→82
Formula 1	484→478	172→127
Superhero	508→501	129→118
Thrombosis prediction	311→304	161→126
Total	3,835→3,778	1,118→906

Each external method supplies one unpaired 2,000-item set. Omni replaces duplicates before response inspection; Synth expands adaptively and deduplicates in fixed order. Synth mixes reused/new outputs and batch/KV settings; 366 tasks lack post-run model checks after interrupted shards. These limitations and unequal generation budgets qualify the comparison.

3) Scoring Settings: We apply both scoring layers from Section IV-C2 to the same selected tasks. Unknown scores fall from 692 to 607 among 10,340 source outputs and from 195 to zero among 9,060 reference outputs. Synth gains 1,433 matches and 1,676 mismatches; Omni gains 18/758, and other populations gain mismatches only. Proof coverage is unequal, concentrated in California for the first four methods. Semantic assessment is separate (Section V-D2).

B. Main Results

1) Source Fidelity and Target Alignment: Source fidelity and target alignment favor different methods (Table II). Schema2Bench improves Template's Verified gap by 0.055693 and source TEF over all comparators; Ring, Omni and Synth have lower gap.

Figure caps mark bounds; ticks mark coincident endpoints.

All four mean Verified TEF contrasts remain negative across ten overlapping releases, reusing fixed Ring/Omni/Synth sets. Base means cross zero for Template/Ring/Synth; Omni remains favorable.

2) Model Ranking: Across ten releases under Verified scoring, mean gap decreases from 0.224640 to 0.168988 (5.5652 pp) and mean domain τ-b rises from 0.2292 to 0.3272. California and codebase nevertheless lose rank fidelity, distinguishing gap magnitude from ordering.

Per seed, 1,000 bootstrap draws separately resample matched SQL-skeleton clusters and references. All ten rank-difference estimates are positive; 8/10 conditional 95% intervals exclude zero, omitting missingness and pool selection.

3) Model Selection: The regret rule was fixed before this post-hoc calculation. It retains all possible winners, including ties and unknowns, without ranking interval midpoints.

With 906 Verified references, Ring has lower mean regret despite S2B's lower TEF (Table III), favoring Ring in California/codebase and S2B in Formula 1/superhero. Omni/Synth

TABLE II

FIXED FIRST RELEASE: 2,000 TASKS/METHOD, TEN MODELS, 906 TARGET REFERENCES AND 1,034 SOURCE SPIDER ITEMS. ↓: LOWER IS BETTER. DIRECT ΔTEF: S2B MINUS ROW METHOD, SHARING SOURCE UNKNOWNS; NEGATIVE FAVORS S2B. UNKNOWNS COUNT CELLS OUT OF 20,000; INTERVALS BOUND UNKNOWN SCORES.

Method	Target gap $G \downarrow$	Spider–BIRD TEF ↓	Direct ΔTEF	Unknown
Verified execution scoring				
Template	$[0.224967, 0.224967]$	$[0.243915, 0.390986]$	$[-0.131730, -0.079099]$	0
S2B	$[0.169275, 0.169275]$	$[0.138501, 0.283651]$	0	0
RingSQL-style adaptation	$[0.096309, 0.100791]$	$[0.283403, 0.435663]$	$[-0.233594, -0.061506]$	46
OmniSQL component adaptation	$[0.029731, 0.134760]$	$[0.564696, 0.862439]$	$[-0.665769, -0.324655]$	1,290
SynthSQL-style local adaptation	$[0.013004, 0.166928]$	$[0.268596, 0.642072]$	$[-0.453605, -0.021084]$	2,040
Base layer: strict execution and output-shape checks				
Template	$[0.178882, 0.262025]$	$[0.205735, 0.426469]$	$[-0.193294, +0.005344]$	430
S2B	$[0.117662, 0.208049]$	$[0.111299, 0.334757]$	0	537
RingSQL-style adaptation	$[0.059199, 0.138379]$	$[0.241426, 0.462380]$	$[-0.293401, +0.034557]$	445
OmniSQL component adaptation	$[0.010465, 0.190895]$	$[0.488896, 0.898945]$	$[-0.734008, -0.195186]$	2,066
SynthSQL-style local adaptation	$[0.000000, 0.345368]$	$[0.103282, 0.861883]$	$[-0.686921, +0.198272]$	5,149

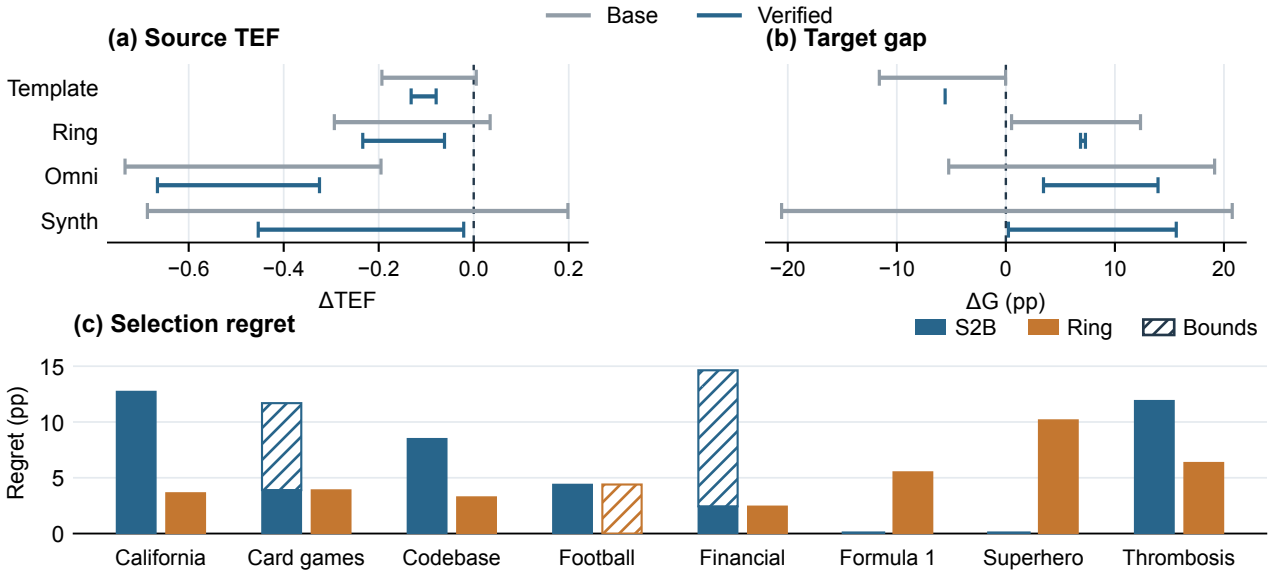


Fig. 3. First-release source TEF, target gap and model-selection regret. Panels (a–b) compare S2B with each method; panel (c) compares S2B and Ring by domain.

remain imprecise. Figure 3(c) separates lower bounds from remaining ranges.

All 1,118 references give overlapping S2B/Ring ranges, with 1,356 Base and 706 Verified unknowns. Across ten overlapping releases with 906 references, S2B's mean lower/upper endpoints span 5.482–7.696/5.482–7.981 pp.

C. Ablation Studies

Selection controls share the candidate pool; column preference and value sampling use separate paired releases; question rewriting holds SQL fixed. Figure 4 separates TEF and gap. Effects do not add up to the main gain.

1) Workload Selection: Most Template–S2B gap improvement precedes selection (Table IV(a)). Adjustment adds only 0.0204 pp over matched random under Verified scoring, positive in 9/10 seeds. The 16 random alternatives/seed match screening, directed stratum-removal counts, swaps and H. Thus structural control adds little target-gap benefit.

Initial mean τ-b is 0.2312/0.3270 for Template/S2B versus 0.2292/0.3272 after adjustment. Structural-minus-matched-random TEF is $[+0.000227, +0.000977]$ with source schema summaries (8/10 unfavorable; 2/10 unresolved) and $[-0.000777, +0.000452]$ with source DDL (2/10 favorable; 8/10 unresolved).

a) Seven additional databases: Earlier seven-model results favor bounded selection in TEF and greedy in structure; all 14 swaps occur in TPC-DS (Table IV(b)). Repair improves yield, but controlled construction yields fewer tasks at higher cost than Ring, without establishing main-study component gains.

TABLE III

MODEL-SELECTION REGRET (ACCURACY PP), RETAINING ALL POSSIBLE TIED WINNERS. MEAN/WORST DENOTE DOMAIN AVERAGE/MAXIMUM; 906 AND 1,118 DENOTE REFERENCE COUNTS. COLUMNS DISTINGUISH SCORING LAYERS; ENDPOINTS ARE CONSERVATIVE.

Method	906: base mean	906: verified mean	906: verified worst	1,118: base mean	1,118: verified mean
Template	[6.138, 8.979]	[6.718, 7.854]	[10.317, 12.727]	[0.556, 19.177]	[2.213, 14.329]
S2B	[4.122, 9.106]	[5.482, 7.981]	[12.727, 14.634]	[0.071, 19.192]	[2.500, 14.417]
Ring	[2.887, 8.100]	[4.409, 4.958]	10.169	[0.484, 18.847]	[1.547, 11.519]
Omni	[0.000, 17.895]	[0.794, 12.246]	[6.349, 19.512]	[0.000, 28.023]	[0.000, 17.857]
Synth	[0.000, 27.258]	[0.000, 19.046]	[0.000, 25.490]	[0.000, 37.369]	[0.000, 23.858]

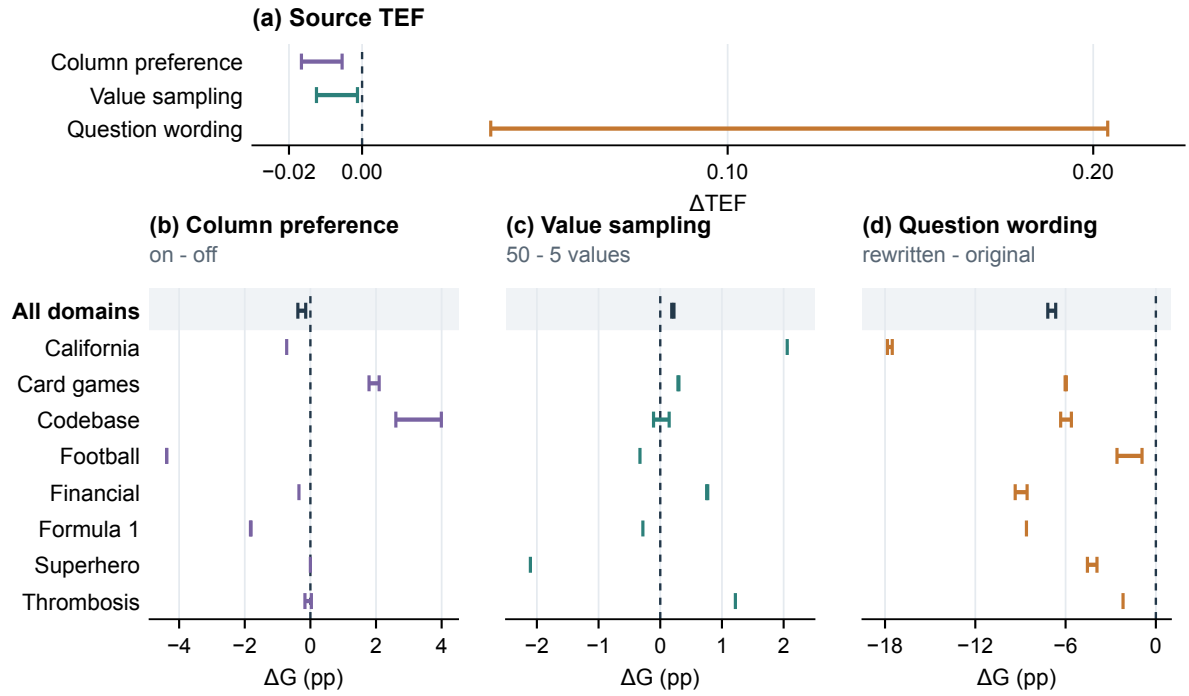


Fig. 4. Construction controls under Verified scoring: source TEF and target-gap differences across domains.

TABLE IV

SELECTION RESULTS: TEN OVERLAPPING 2,000-TASK RELEASES PER STUDY. (A) BIRD, TEN MODELS; POSITIVE $I_G = G(\text{Template}) - G(\text{S2B})$ FAVORS S2B. (B) EARLIER SEVEN-DATABASE, SEVEN-MODEL STUDY; LOWER LOSSES ARE BETTER.

(a) BIRD: construction versus selection			
Selection	Swaps	H	I_G (pp)
Initial stratified sample	0	0	5.5386
Matched random swaps	10	0.01	5.5449
Structural adjustment	10	0.01	5.5652
(b) Seven additional databases: selection			
Selector	Structural loss		TEF
Stratified random	0.269519		0.050071
Structural greedy	0.213055		0.048692
Bounded selection	0.260790		0.047229

2) Column Preference and Value Sampling: Separate releases disable descriptive-column preference or reduce value sampling from fifty to five. They share the main model/source/reference panel, 38 strata, 250 tasks/domain/arm and $H = 0.01$. Attempts pair random states and plans; separately deduplicated candidates must qualify in the same final stratum.

The preference control removes only priority for descriptive column names, retaining identifier filtering, text preference, grounding and admission. Programmatic question generation uses eight-second and 1,000-row caps. Four domains require 1,600 extra slots each before freezing; the arms share 1,159/2,000 question–SQL slots. Depth fixes 2,400 slots/domain/arm: fifty/five values admit 7,131/6,149 candidates, with 5,375 joint admissions and 769 shared released slots. Astra supports both versions of 80 depth pairs, 24 with identical prompts, using exposed controls. These admission differences remain part of the controls.

SQLite compilation resolves 827 preference and 903 depth parser rejections as mismatches without rewriting SQL. Identical requests and outputs share unknown outcomes across arms; other unknowns remain relaxed. This post-hoc treatment

preserves scores, membership and identity records. Table V retains earlier independent-arm bounds.

Column preference improves both aggregate metrics; deeper sampling trades a small gap cost for better TEF (Table V). Base TEF favors both, with unresolved gaps. LP checks reproduce 26 aggregate/domain contrasts per intervention.

Column preference worsens gap in card games and codebase (Fig. 4(b)). Excluding European football changes its seven-domain mean to $[+0.001899, +0.004625]$; excluding Formula 1 leaves it unresolved. Value-depth effects also vary by domain.

a) Testing the value-depth effect on more databases: Four SynSQL-2.5M training databases [7] supply 1,000 identical pairs; BIRD `student_club` changes 150/250 pairs. Three further targets stop below quota. The combined thirteen-domain analysis uses 3,250 tasks/arm and 1,733 references: Verified TEF/gap differences are $[-0.050927, +0.049902]$/$[-0.029026, +0.030559]$. Adding SQLite evidence only on the original eight domains gives $[-0.013017, +0.011489]$/$[-0.004886, +0.007287]$. Both remain unresolved. Unchanged task pairs, unequal scoring evidence and synthetic/training overlap limit independent replication.

3) Question Generation: To isolate the role of wording, we hold SQL fixed. A preliminary comparison finds 1,973 Ring–Template pairs sharing SQL, domain and schema prefixes but differing in questions. Ring's execution-match rate is $[-13.96, -13.73]$ pp lower, motivating a controlled wording intervention.

We retain all 2,000 first-release S2B SQLs and quotas and use Ring-style Qwen2.5-7B question generation. The plan and 160-item audit (20/domain) precede generation. It yields 1,999 format-valid questions and 19,990 predictions; the remaining item stays unknown.

Rewriting improves Verified gap in all eight domains and aggregate Base gap, but worsens Verified TEF (Table VI). Concordances increase; regret intervals overlap. Against Ring, TEF is lower and gap unresolved.

Terra supports 117/160 tasks using exposed synthetic controls; concerns about changes in task meaning remain despite fixed SQL (Section V-D3). This intervention does not alter the original qualification outcome.

4) Candidate Coverage: The 3,778-pair pool contains 311 easy, 2,688 medium and 779 hard tasks versus source proportions $(0.464, 0.348, 0.188)$. With 38 quotas allowing at most 173 easy items per 2,000, $h = (1, 0, 0)$ certifies $0.464 - 173/2000 = 0.3775$. Both $(173, 1302, 525)$ and the current $(173, 1376, 451)$ attain it. The latter meets $H = 0.01$, so selection cannot reduce difficulty TV. Exact matching requires at least $928 - 173 = 755$ additional eligible easy items, without guaranteeing simultaneous type feasibility.

5) Additional Candidate Generation: On a SynSQL-2.5M garment database with 159 synthetic references, adaptive expansion follows a 223/250 quota failure. Equal 1,200-attempt supplements yield 119 guided versus 29 untargeted candidates, but all pools retain difficulty-TV optimum 0.356. This bound relaxes H and does not bound TEF.

Ten overlapping releases reuse seven BIRD domains, with 250 tasks/domain, $H \leq 0.01$ and identical structural losses. Mean guided-minus-untargeted TEF is $[-0.000294, +0.000334]$; new-target gap is $[-0.000905, +0.000542]$, favorable/adverse/unresolved in 3/1/6 releases. References retain 111/1,590 unknowns. Higher yield leaves measurement contrasts unresolved and the structural deficit unchanged.

D. Robustness Analysis

Each sensitivity analysis specifies its population and uncertainty model.

1) Sensitivity to Sampling and References: For the first-release gap difference -0.0556925, 1,000 bootstrap draws resample paired tasks within domain/stratum and references within domain, retaining joint model responses. The conditional 95% interval is $[-0.0665464, -0.0434725]$.

Leave-one-domain-out contrasts and all 56 three-of-eight means retain the favorable direction. These checks condition on observed pools and omit independent construction variance.

a) A revised target reference: We evaluate all 417 same-domain items in BIRD's October 2026 Mini-Dev SQLite revision.[2] Question and database hashes match the publisher manifest and original snapshots; references do not overlap by domain/question/evidence identity. All SQL executes; 67 predictions truncate.

Revised-reference gap favors S2B over Template in six domains, with California/Formula 1 unresolved (Table IX); S2B-minus-Ring is $[+0.040718, +0.103446]$. The unpaired revision has more tasks rated hard by the SQL complexity measure and more nested SELECTs; domains and source TEF are unchanged.

2) Semantic Assessment: OpenAI's `gpt-6-astra` API judges question–SQL agreement [24, 25] from question, schema and SQL with a fixed prompt and medium reasoning, blind to methods/scores. Qwen/Terra cross-check supported/incorrect/insufficient labels. Evaluator biases [26] preclude treating these as human truth or calibrated confidence; sensitivity relaxes flagged scores at fixed denominators.

a) Generated-task audits: An earlier stratified 400-task audit also leaves Synth unresolved under Terra's 135/137-concern scenarios. Retaining only 654 nonempty Synth roundtrip matches gives ten-release mean TEF bounds $[-1.076911, +0.283264]$; other scores are unknown at the fixed denominator.

The expanded audit fixes 100 records per method/domain, with one unavailable (Table VIII); exposed controls score 24/24 per class. The union of 1,133 audit concerns and 620 earlier SQL flags relaxes 1,433 items at the original denominators.

Relaxing all 1,433 flags preserves S2B's TEF advantage over Template, $[-0.179682, -0.050176]$, but leaves external contrasts unresolved. Shared-source LP bounds for Ring/Omni/Synth are $[-0.293390, +0.139461]$, $[-0.840707, +0.132174]$ and $[-0.587412, +0.236106]$; joint

[2]https://github.com/bird-bench/mini_dev

TABLE V

SEPARATE CONSTRUCTION CONTROLS UNDER VERIFIED SCORING. NEGATIVE FAVORS FULL PREFERENCE OR FIFTY VALUES. BOUNDS WITH SHARED VARIABLES BIND IDENTICAL OUTCOMES; INDEPENDENT-ARM BOUNDS RETAIN THE EARLIER RELAXATION.

Intervention	Unknown-variable treatment	Direct TEF difference	Direct target-gap difference
Column preference	Independent arms	$[-0.029752, +0.008062]$	$[-0.011942, +0.006559]$
Column preference	Shared identical requests	$[-0.016593, -0.005466]$	$[-0.003816, -0.001425]$
50 versus 5 values	Independent arms	$[-0.024987, +0.011333]$	$[-0.006878, +0.011031]$
50 versus 5 values	Shared identical requests	$[-0.012477, -0.001261]$	$[+0.001863, +0.002221]$

TABLE VI

QUESTION REWRITING WITH FIXED SQL ON THE FIRST RELEASE. PANEL (A): VERIFIED OUTCOMES; CONCORDANCES ARE DEFINITE MATCHING ORDERS AMONG 360 DEPENDENT MODEL PAIRS; REGRET IS IN ACCURACY PP. PANEL (B) REPORTS DIRECT BOUNDS UNDER UNKNOWN SCORES; NEGATIVE FAVORS REWRITING.

(a) Overall results Release	Target gap	Definite concordances	Mean regret (pp)
Original S2B	0.169275	229/360	$[5.48, 7.98]$
Rewritten S2B	$[0.097452, 0.102706]$	248/360	$[4.30, 5.57]$
Ring	$[0.096309, 0.100791]$	246/360	$[4.41, 4.96]$
(b) Direct differences Contrast	Scoring	ΔTEF	ΔG
Rewritten − original S2B	Verified	$[+0.035191, +0.203968]$	$[-0.071823, -0.066569]$
Rewritten − Ring	Verified	$[-0.046174, -0.009796]$	$[-0.003339, +0.006397]$
Rewritten − original S2B	Base	Crosses zero	$[-0.120114, -0.002971]$

TABLE VII

RECORDED TOKENS INCLUDING FAILURES. EVALUATION ROWS ARE RELEASE MEANS; GENERATION ROWS COVER DIFFERENT SCOPES.

Stage	Input tokens	Output tokens	Scope
Template evaluation	22,906,504	1,170,824	20,000 model–item records/release
S2B evaluation	22,936,877	1,143,564	20,000 model–item records/release
Spider summary evaluation	1,816,236	370,097	10,340 source records
Spider DDL evaluation	4,221,154	370,682	10,340 source records
Synth final evaluation	22,041,137	804,975	20,000 records; 5,080 reused
Ring question generation	417,101	125,758	2,400 records; excludes template SQL
Omni SQL and question generation	31,498,785	5,905,818	Initial, additional and reserve generation
Synth all recorded generation	553,507,001	15,411,464	258,674 calls, including failed/stopped attempts

gap bounds also cross zero. Three of 48 LP/mixed-integer gap endpoints reach time limits. Outward tolerances and earlier wider bounds remain.

b) Target-reference audits: Accepting JSON inside code fences gives Qwen/Terra agreement on 107/160 references (Table VIII(b)); independently generated SQL sometimes contradicts the labels. The 105-item strict union treats format failures and non-supported judgments as risks, not established errors.

The 71-reference scenario favors S2B in all ten releases. Combining strict cross-check or revised references with generated risks leaves the first release unresolved (Table IX). Reference-only changes leave TEF unchanged.

3) Sensitivity to Question Rewriting: Figure 5 separates construction (43 rewritten, seven original S2B, 23 Ring) and reference risks (71 Terra or 105 strict Terra/Qwen). Each flag relaxes all ten scores, fixing others and sharing source/reference variables. "Both" combines flags, excluding the separate 1,433-item union.

Construction flags preserve the rewrite's gap advantage but leave both TEF contrasts unresolved. Either reference-risk set makes gap direction unresolved without changing TEF.

Positive differences are feasible in 14/16 Verified cases whose bounds cross zero; this establishes neither semantic errors nor reversal probabilities.

Terra reviews 160 paired originals under the same protocol (Table VIII(c)). Combined flags cover 24 original, 43 rewritten, 23 Ring and 71 reference items. Verified rewrite-minus-original gap $[-0.142130, +0.032135]$ and TEF $[-0.008539, +0.235566]$ are both unresolved.

4) Metric Weights and Model Coverage: Accuracy and the correct-count CDF favor S2B against all comparators; disagreement favors it over Template, is adverse to Ring ($[+0.012123, +0.066481]$), and unresolved for Omni/Synth. Equal component weights and pooled-task averaging leave S2B-minus-Synth unresolved: $[-0.400081, +0.034253]$ and

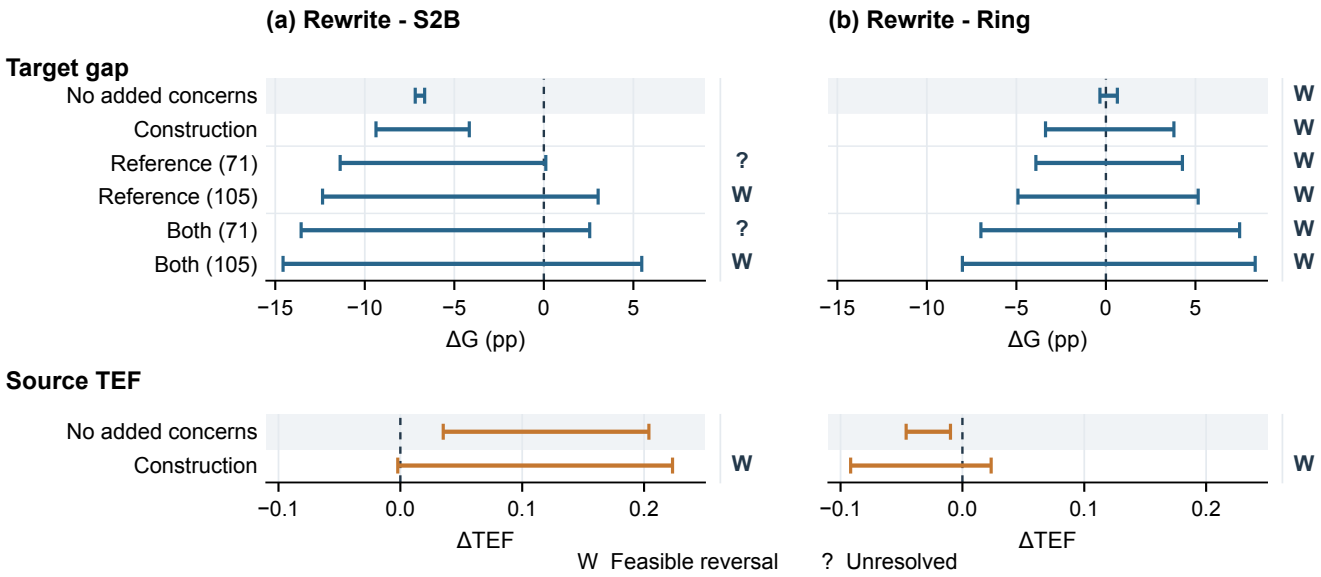


Fig. 5. Sensitivity of rewriting with fixed SQL using earlier audit flags. Target gap is above source TEF. W: a feasible score assignment reverses the comparison; ?: search unresolved; blank: not searched.

TABLE VIII
API JUDGMENTS. S/I/U: SUPPORTED/INCORRECT/INSUFFICIENT; NA/INV.: UNAVAILABLE/INVALID. PANELS (B)/(C) COMPARE JUDGES ON SHARED REFERENCES AND PAIRED QUESTIONS UNDER THE SAME TERRA PROTOCOL.

(a) Astra: 800 tasks/method, 100/domain				
Method	S	I	U	NA/inv.
Template	800	0	0	0
S2B	732	17	51	0
Ring	603	123	73	1
Omni	290	418	92	0
Synth	442	308	50	0
(b) References: 160 tasks, 20/domain				
Judge / parsing rule	S	I	U	NA/inv.
Terra	89	54	17	0
Qwen: strict JSON	76	9	1	74
Qwen: code-fence parsing	121	37	2	0
(c) Terra: 160 paired tasks, 20/domain				
Original S2B	141	14	5	0
Rewritten S2B	117	32	10	1

$[-0.430144, +0.025669]$. Template/Ring/Omni directions persist.

Figure 6 varies nonnegative accuracy/CDF/disagreement weights summing to 2.5 on the first Verified release without semantic flags. Colors classify regions as favorable, adverse or unresolved using conservative bounds, without assigning probabilities.

S2B beats Template throughout the weight region. Emphasizing disagreement can favor Ring and leaves Omni/Synth unresolved.

Removing Qwen retains favorable Verified TEF contrasts after recomputing normalizers; external adaptations still have lower gap (Table X). Summary/DDL source prompts both favor S2B over Template, with unresolved direct difference.

Broader coverage retains the Template–S2B gap advantage (Table X). Adding Qwen3.6-27B, Qwen3.8-27B, Gemma 3 4B and OLMo 2 7B tests model coverage, without resolving exclusion bias, transfer or training overlap.

LIMIT counts are 1,739/2,000 for Template/S2B, 1,753 Ring, 424 Omni and 872 Synth, versus 62/906 references. Only 108/240 method–domain–structure cells (10/40 LIMIT cells) have ten tasks in both populations, preventing complete matched analysis. A pilot table-name repair does not apply to the retained failed outputs.

E. Generalization Experiments

TACO OpenData [27] supplies two related Beijing databases and Connecticut, with potentially synthetic references. Admission and initial sampling differ from the main study. With 16 retrieved objects, contrasts are unresolved against Template and adverse against Ring (Table XI). Incomplete schemas limit semantic assessment; Omni/Synth miss 250-task quotas with incomplete runs.

A ten-model pilot uses CORE-T's SQLite-converted BEAVER [28].[3] DW/Nova/Neutron supply unequal capacities of 250/198/24 tasks/method and all 121/43/30 references; Nova/Neutron share provenance. Retrieval covers required objects for 33/194 references. Oracle schemas use reference-SQL-selected tables/views and compact source DDL, hiding SQL. This tests known relevant schemas, changing coverage and input length; it does not test retrieval.

In both Base conditions, Schema2Bench's pooled TEF advantage over Template persists; against Ring it crosses

[3]https://github.com/UKPLab/emnlp2026-core-t

TABLE IX
S2B-MINUS-TEMPLATE GAP SENSITIVITIES ON THE FIRST VERIFIED RELEASE. EACH RISK SET RETAINS ITS DENOMINATOR AND UNFLAGGED SCORES; RANGES BOUND UNKNOWN SCORES.

Generated-task concerns	Reference condition	Reference count	Direct gap difference
None added	Original reference	906	-0.055693
1,433 union flags	Original reference	906	$[-0.077344, -0.026784]$
None added	71 Terra reference concerns	906	$[-0.098374, -0.010170]$
137 original audit flags	105 strict Terra/Qwen concerns	906	$[-0.114491, +0.013951]$
137 original audit flags	79 code-fence sensitivity concerns	906	$[-0.105686, -0.002284]$
None added	Revised BIRD reference	417	$[-0.076995, -0.024912]$
1,433 union flags	Revised BIRD reference	417	$[-0.105428, +0.000213]$

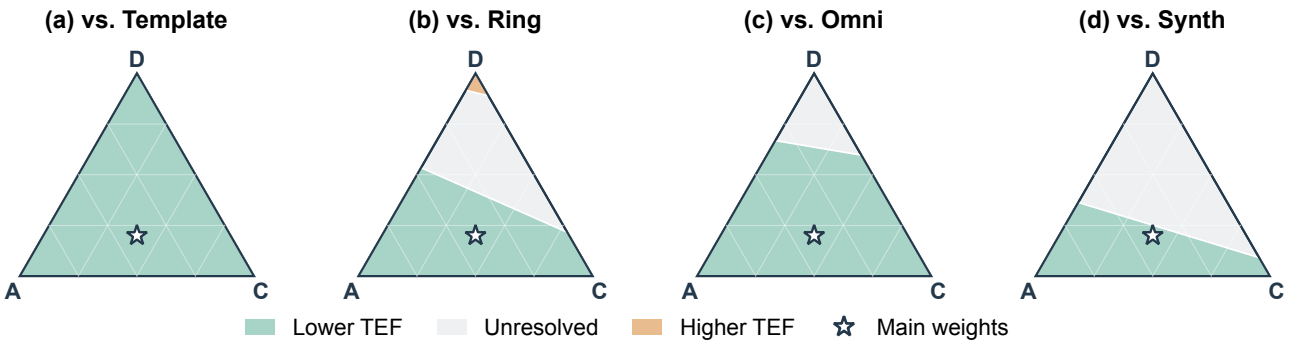


Fig. 6. Post-hoc TEF weight sensitivity. A: accuracy; C: correct-count CDF; D: disagreement. Star: original weights $(1, 1, 0.5)$.

TABLE X
EIGHT-DATABASE GAP SENSITIVITY: POSITIVE $I_G = G(\text{Template}) - G(\text{S2B})$ FAVORS S2B. CONDITIONAL CONFIDENCE INTERVALS (CIS) RESAMPLE WITHIN DOMAINS; BOUNDS DESCRIBE UNKNOWN SCORES.

Setting	Pairs	References	Gap improvement I_G	Interpretation
Non-Qwen panel	2,000	906	0.026560	Ten-release mean
Broader reference coverage	3,778	982	0.053302 $[0.041870, 0.063207]$	Conditional 95% CI
Equal-skeleton weighting	3,778	982	0.046936 $[0.034298, 0.057670]$	Conditional 95% CI
Full available pool	3,835	1,118	$[0.009303, 0.090704]$	Unknown-score bound
Fourteen models: strict	2,000	906	$[0.002662, 0.059226]$	Unknown-score bound
Fourteen models: database checks	2,000	906	$[0.025795, 0.032495]$	Unknown-score bound
Fourteen models: strict	3,778	906	$[0.002188, 0.059640]$	Unknown-score bound
Fourteen models: database checks	3,778	906	$[0.025787, 0.032648]$	Unknown-score bound

The full-pool setting uses 8,788 prompts/model. The 2,000-pair fourteen-model rows average bound endpoints over ten overlapping releases. Fourteen-model concordances/inversions are 567/136 for 2,000 pairs and 533/168 for 3,778 pairs, each out of 728 dependent model pairs.

zero with oracle schemas. Neutron's oracle TEF/gap favor Ring. Reference unknowns remain 660/1,940 and 663/1,940. Unresolved gap and wide rank/regret bounds do not establish better target model selection.

a) Seven-database diagnostic: A separate ten-model diagnostic on the seven earlier databases finds no general Schema2Bench advantage. Equal-domain means of Verified direct TEF bounds are $[-0.1055, +0.0750]$, $[+0.0206, +0.2190]$, $[-0.5911, +0.1033]$ and $[-0.4155, +0.4507]$ for S2B minus Template, Ring, Omni and Synth fallback. These average domain contrasts, not pooled tasks. S2B beats Ring on TPC-H, loses on Chinook, Instacart, SFDFFD and TPC-DS, and is unresolved on IMDb/USAccidents.

Task counts vary: TPC-DS Omni uses 8,192 tokens; TPC-H Omni retains partial outputs; Synth uses fallback generation. Original qualification results remain unchanged. Independent target references are unverified, precluding gap, rank and regret claims.

F. Discussion

Ring has lower gap/regret despite worse source fidelity and fewer Astra-supported pairs (Table VIII). Fixed-SQL rewriting also changes this tradeoff: structural coverage, semantic support and model choice require separate checks.

An illustrative 0.003 tolerance (0.3 pp) accepts deeper sampling's gap cost of at most 0.002221 for better TEF, but not Rewritten-minus-Ring's 0.006397 upper bound. Prospective

TABLE XI
EXTERNAL-PILOT DIFFERENCES (S2B MINUS COMPARATOR); NEGATIVE FAVORS S2B. TACO: 750 TASKS/METHOD, 595 REFERENCES, DATABASE AND REFERENCE CHECKS. BEAVER: 472 TASKS/METHOD, ALL 194 REFERENCES. SCORING LAYERS AND SCHEMAS ARE SEPARATED. EACH PILOT USES MATCHING SOURCE CONTEXT, POOLED TEF AND GAP AVERAGED ACROSS DOMAINS.

Setting	Comparator	ΔTEF	ΔG
TACO: retrieved schema	Template	$[-0.006024, +0.016398]$	$[-0.030517, +0.044555]$
TACO: retrieved schema	Ring	$[+0.133673, +0.280760]$	$[+0.148866, +0.277121]$
BEAVER: retrieved, Base	Template	$[-0.146373, -0.033660]$	$[-0.081035, +0.060364]$
BEAVER: retrieved, Base	Ring	$[-0.242054, -0.013075]$	$[-0.056789, +0.099330]$
BEAVER: oracle schema, Base	Template	$[-0.142702, -0.028519]$	$[-0.083884, +0.049262]$
BEAVER: oracle schema, Base	Ring	$[-0.130602, +0.168975]$	$[-0.051570, +0.064633]$
BEAVER: retrieved, Verified	Template	$[-0.146163, -0.037765]$	$[-0.081035, +0.060364]$
BEAVER: retrieved, Verified	Ring	$[-0.235815, -0.016203]$	$[-0.056789, +0.099330]$

non-inferiority tests require fixed tolerances and sufficient precision.

Screening, unequal budgets and evidence, shared data and possible training overlap limit generalization. External results are mixed; risk bounds keep unflagged scores fixed.

Table VII includes failed-call tokens. Retained artifacts support offline checks of scores, feasibility and saved execution evidence. Full reproduction requires external models, data and environments; redistribution rights remain unverified.

VI. RELATED WORK

A. Text-to-SQL

Text-to-SQL research has improved schema linking, representation learning and constrained decoding [29–33]. DAIL-SQL studies prompting and example selection, and CodeS develops specialized open models [1, 2]. PURPLE retrieves demonstrations with relevant operator compositions; KaSLA formulates schema linking as knapsack optimization [34, 35]. Long-context NL2SQL examines the accuracy and latency effects of schema, values and other context [36]. These methods improve predictors; Schema2Bench constructs workloads for comparing them on new databases.

DIN-SQL, MAC-SQL and SHARE use decomposition, collaboration or correction [37–39]; CycleSQL uses database-grounded explanations as feedback [3]. These checks improve predictions but do not establish benchmark fidelity. Execution agreement itself requires qualification through test suites, formal checks and reference audits [9, 40–42].

Spider, BIRD and conversational benchmarks provide shared evaluation tasks [4, 5, 43, 44]. Spider 2.0, KaggleDBQA, ScienceBenchmark, BEAVER and customer-log evaluations extend the range of database demands [28, 45–48]. Perturbation benchmarks isolate linguistic, knowledge, schema and query changes [11, 49–53]. SQLMorph adds join expansion and fine-grained execution metrics; VLD-Bench studies discrepancies between question values and database entries [8, 10].

These diagnostics isolate prediction failures under specified task changes. They do not establish whether migration preserves a source response profile while reflecting model differences on a separate target workload. Following evidence-centered design and concurrence research [12, 13, 54], we separately evaluate source response profiles and target accuracy gaps, rankings and regret, retaining uncertainty when execution or semantic evidence is insufficient.

B. Data Synthesis

Synthetic question–SQL pairs support pretraining and adaptation [55–57]. RingSQL instantiates schema-independent templates; OmniSQL synthesizes large-scale training data [6, 7]. SING-SQL, EvolSQL and SynSQL study other synthesis settings [58–60]. These training objectives do not establish evaluation fidelity. We compare local adaptations at fixed workload sizes and isolate wording effects with SQL fixed.

Dataset distillation optimizes a small synthetic training set to approximate learning from a larger one [61, 62]. Gradient-matching condensation explicitly matches training gradients. Schema2Bench constructs test workloads across databases without updating evaluator parameters. A synthetic training set is judged by the model learned from it; a migrated test set is judged by the model comparisons it supports. These objectives require different validation.

Data selection reduces training or evaluation cost [63–65]. TinyBenchmarks estimates full-benchmark performance from selected items; SMART filters items; SubLIME and FAFI study rank preservation [14, 66–68]. Schema2Bench uses moment matching and benefit–cost exchanges [69–71] to control SQL composition under domain quotas. Unlike reducing an existing test set, migration changes both database grounding and the candidate population. The framework bounds fidelity changes under structural exchanges and diagnoses candidate shortages that selection cannot fix. The small additional selection gain and mixed external results delimit the empirical benefits.

VII. CONCLUSION

Schema2Bench combines Benchmark Construction, Workload Selection and Benchmark Evaluation. On eight BIRD databases, it improves source fidelity under Verified scoring and declared weights; external adaptations have lower target gap. Most improvement precedes selection, while fixed-SQL rewriting changes the tradeoff. Semantic uncertainty and mixed external results limit generalization, requiring separate assessment of source continuity and target model choice.

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